

Assessing the manipulative potentials of monkeys, apes and humans from
hand proportions: Implications for hand evolution

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Electronic Supplementary Information

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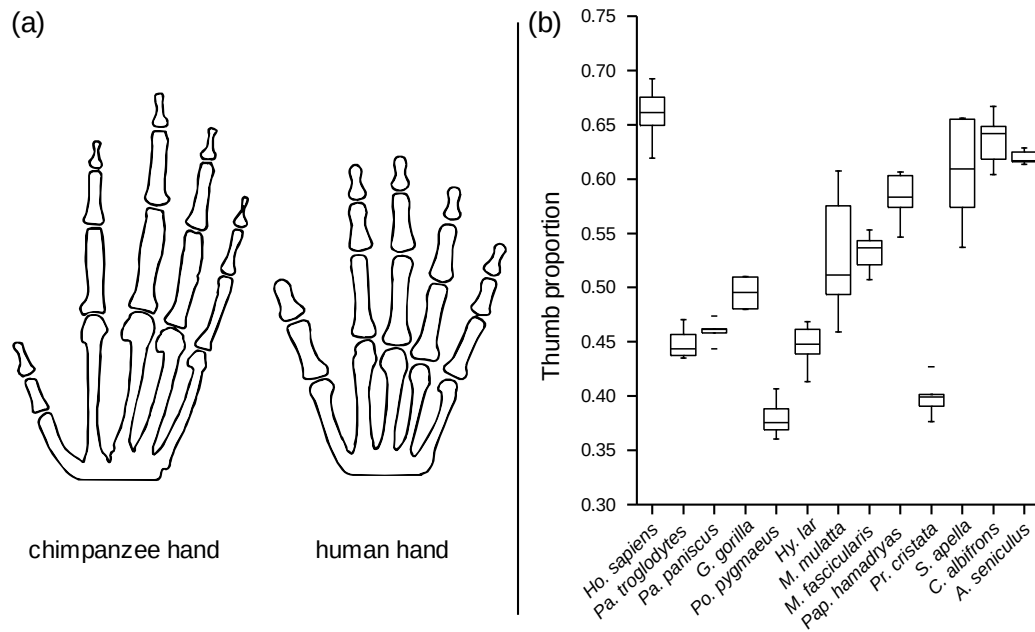


Figure S1. Thumb proportion of the anthropoid hands. (a) The skeleton drawings of chimpanzee hand and human hand in the standardized scale. A most obvious difference between them is the length ratio between the thumb and fingers. (b) The thumb proportion among different anthropoids. The length ratio is calculated by the overall length of the metacarpal, proximal and distal phalanges of thumb dividing the overall length of metacarpal, proximal, intermediate and distal phalanges of forefinger.

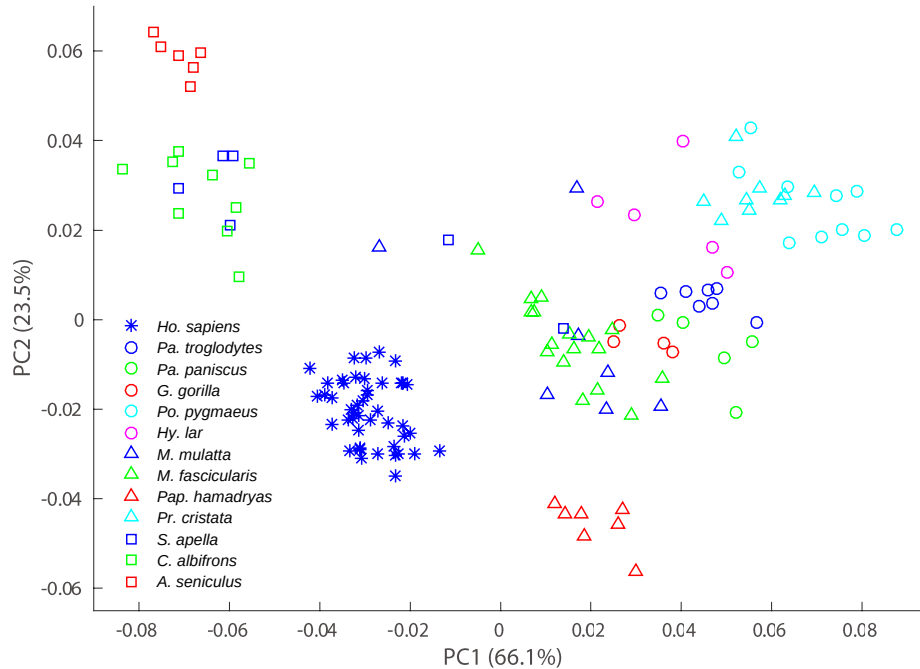


Figure S2. Scatterplot showing first and second principal components for the hand proportions among our anthropoid samples. Individual specimens are plotted so as to better represent the full range of intraspecific variation.

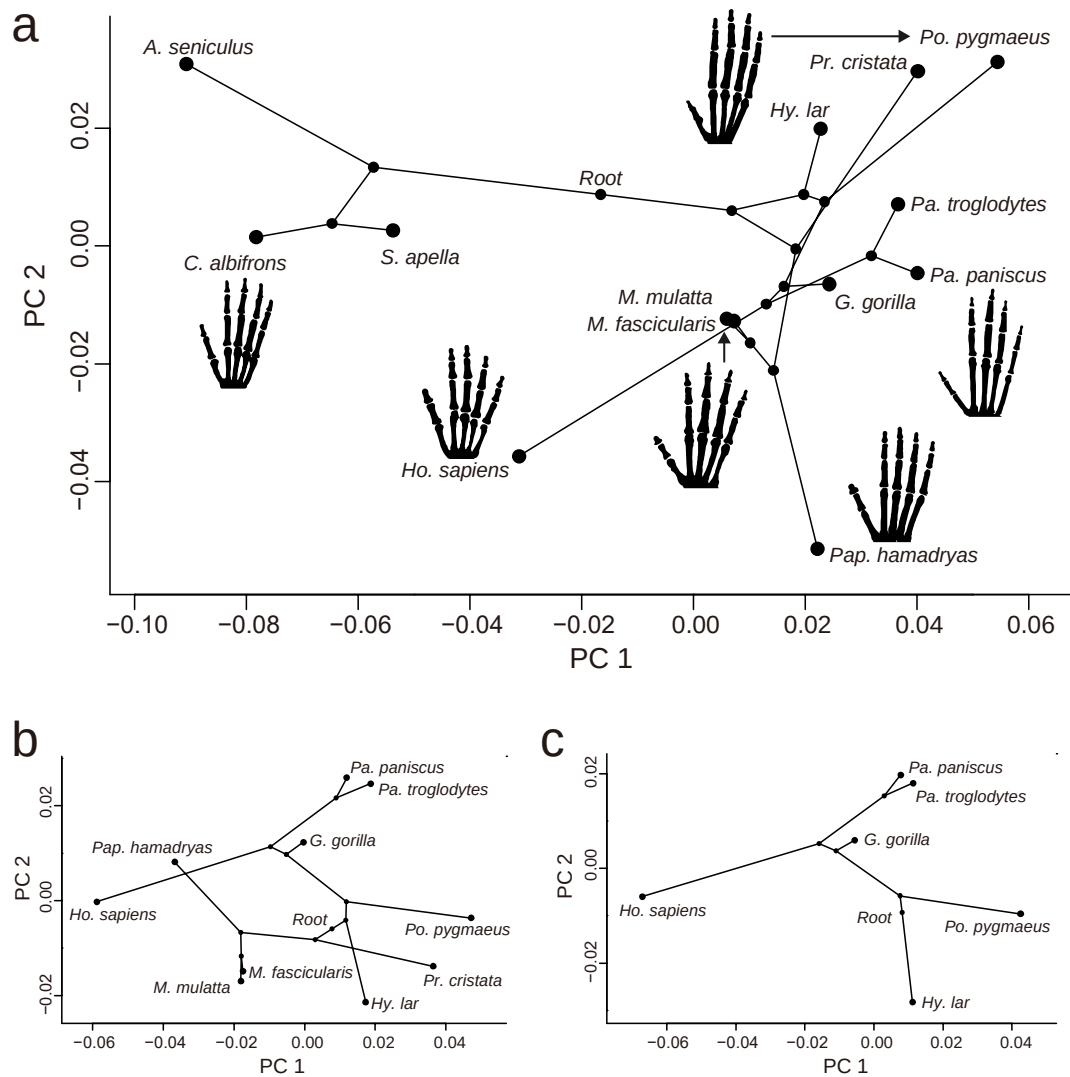


Figure S3. Phylogenetic signal of hand proportions in the anthropoids. The phylogenetic tree is mapped into the morphospaces defined by the first two principal components of different species: (a) all our anthropoid species, $K=1.16$, $P=0.004$; (b) catarrhines only, $K=0.55$, $P=0.168$; (c) hominoids only, $K=0.67$, $P=0.424$, where the statistic K gauges the amount of phylogenetic signal and the P value represents the statistical significance of the observed phylogenetic signal. The high value of K implies the high degree of phylogenetic structure, which means the hand proportions of close relatives more resemble each other than the distantly related species. The evolutionary process of hand proportions would satisfy an expected phylogenetic structure when Statistic- K reaches to one. The statistical significance is set as 0.05. The result indicates that the phylogenetic structure is only found in the case of all anthropoids. The absent degree of phylogenetic signal is especially large in the hominoids.

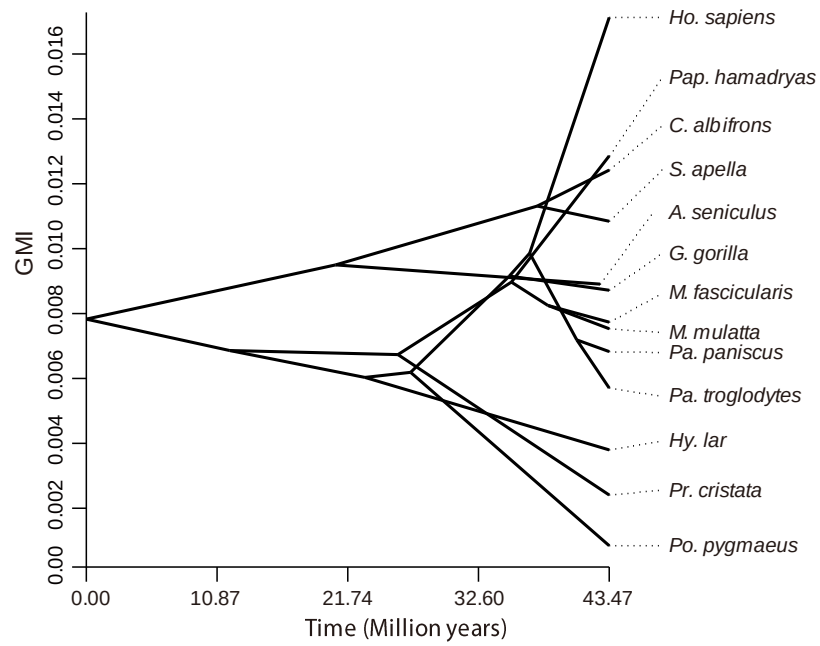


Figure S4. Phenogram plot of manipulative potentials among the anthropoid species. It is a projection of the phylogenetic tree into the space defined by the manipulative potential and time. The manipulative potentials of the anthropoid hands with different segment proportions are assessed by our functional model using the index of GMI. The ancestral states in the internal nodes are estimated by maximum likelihood.

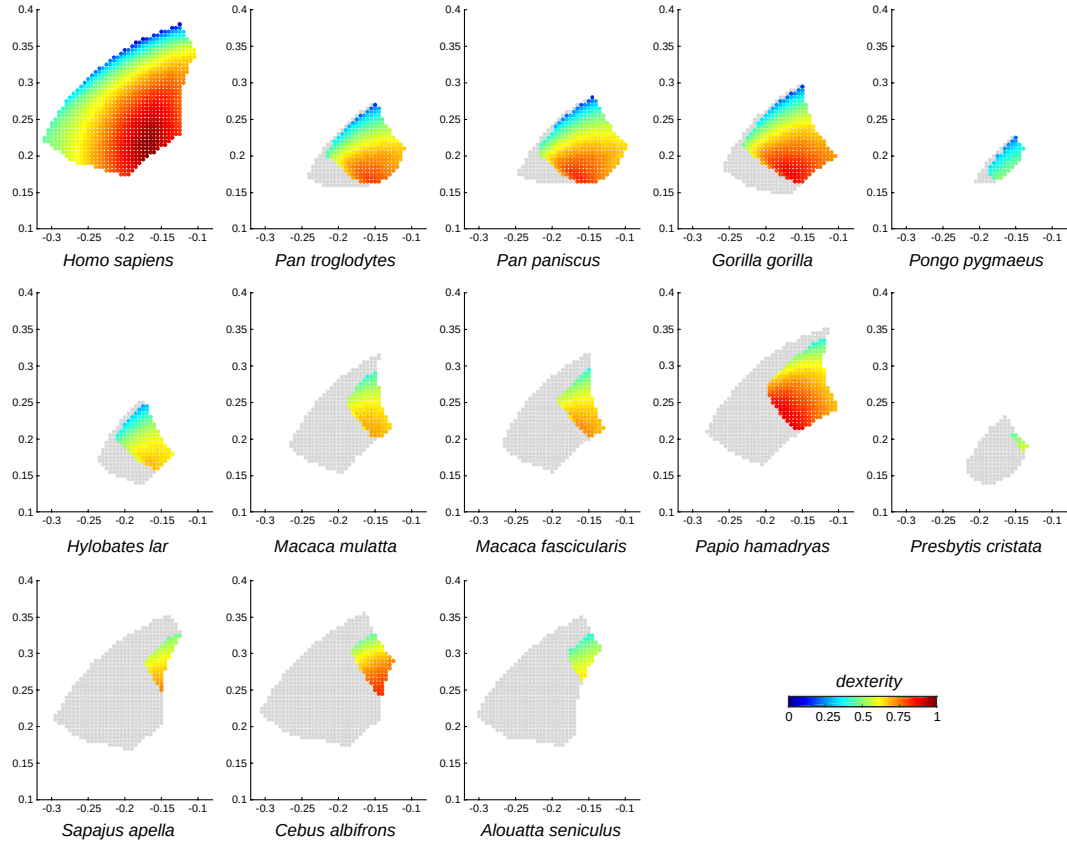


Figure S5. Effect of the constraints of joint mobility on manipulative potential. Compared with the manipulative potential assessed by hand proportions in figure 3, the gray background in each subplot illustrates the lost reachable area due to the constraint of range of motion.

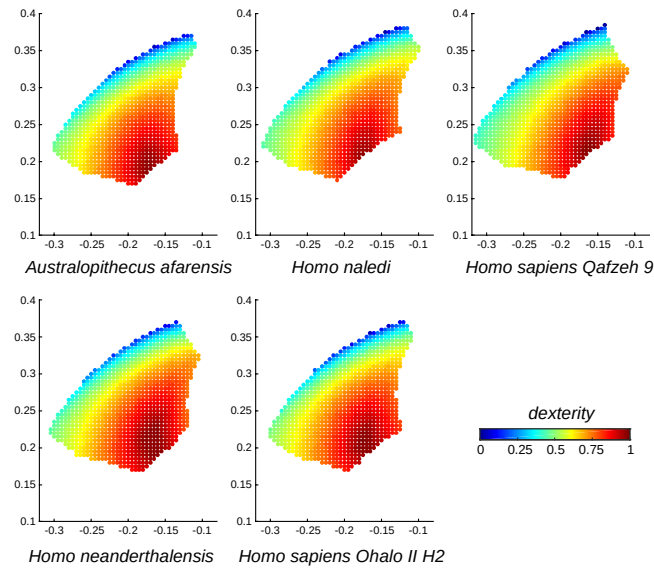


Figure S6. Inference of manipulative potentials in the fossil hands. The morphometrics of these fossil hands are taken from the literature [1], except that of *Homo naledi* fossil hand from the literature [2].

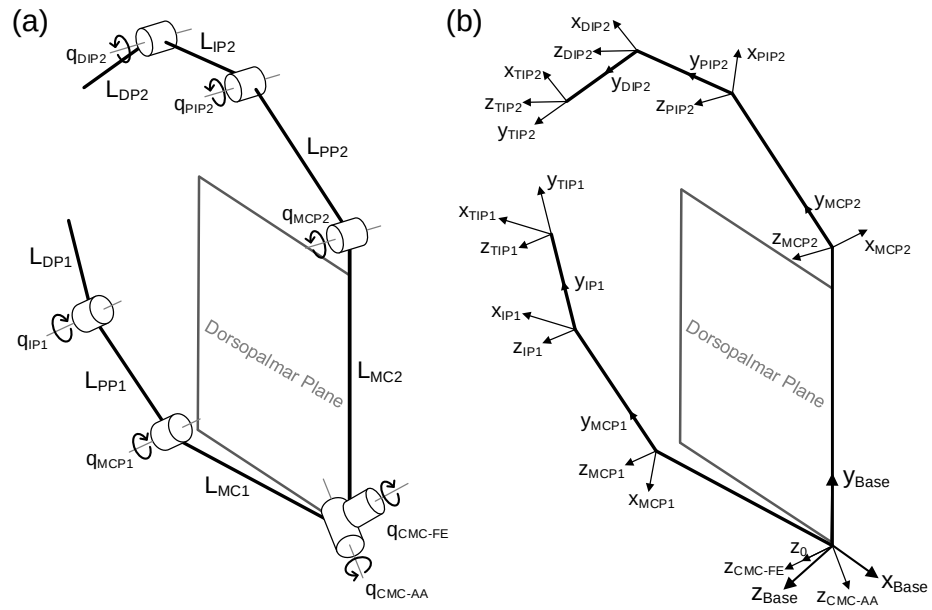


Figure S7. The kinematic joints of the thumb and forefinger (a) and corresponding coordinate systems (b). The local coordinate systems are attached to the rotated joints, with the z axis aligned with the direction of rotation. The abbreviations of the joint and skeletal names refer to those in figure 2. L_{MC1} , L_{PP1} , L_{DP1} , L_{MC2} , L_{PP2} , L_{IP2} and L_{DP2} represent the lengths of the respective finger segments.

Table S1. Hand proportions of the anthropoid samples and their predicted results of manipulative potentials. The length of each hand segment in the samples is the proportion relative to the overall length of thumb and forefinger of associated hand skeletons. The raw morphometrics of these samples are taken from the literature [1]. The museum keys are as follows: AMNH, American Museum of Natural History New York; HMNH, Harvard Museum of Natural History; YPM, Yale Peabody Museum; UM-APC, University of Massachusetts Amherst; ZMB, Museum für Naturkunde Berlin; NMW, Naturhistorisches Museum Wien; UV, University of Vienna; UNI-FI, University of Florence; MRAC, Musée royal de L’Afrique central; NME, National Museum of Ethiopia; WITS, University of this Witwatersrand. Abbreviation: MC1, metacarpal of thumb; PP1, proximal phalanx of thumb; DP1, distal phalanx of thumb; MC2, metacarpal of forefinger; PP2, proximal phalanx of forefinger; IP2, intermediate phalanx of forefinger; DP2, distal phalanx of forefinger; WS, workspace; GMI, global manipulation index.

Species	Key	MC1	PP1	DP1	MC2	PP2	IP2	DP2	WS	GMI
<i>Homo sapiens</i>	AMNH 204072	0.191	0.120	0.087	0.275	0.162	0.099	0.066	0.0250	0.0171
	AMNH 172124	0.174	0.122	0.092	0.268	0.169	0.101	0.074	0.0236	0.0161
	WITS 1301	0.180	0.127	0.087	0.277	0.165	0.094	0.071	0.0246	0.0167
	WITS 1408	0.180	0.133	0.088	0.263	0.174	0.100	0.062	0.0244	0.0155
	WITS 1501	0.171	0.128	0.094	0.262	0.165	0.101	0.079	0.0245	0.0167
	WITS 1576	0.180	0.128	0.092	0.265	0.159	0.100	0.077	0.0257	0.0177
	WITS 1689	0.189	0.127	0.089	0.277	0.155	0.097	0.067	0.0267	0.0182
	WITS 1791	0.184	0.123	0.098	0.258	0.168	0.100	0.070	0.0267	0.0177
	WITS 1918	0.184	0.124	0.096	0.276	0.164	0.092	0.064	0.0273	0.0183
	WITS 1925	0.167	0.125	0.090	0.270	0.173	0.100	0.074	0.0224	0.0149
	WITS 1947	0.175	0.124	0.097	0.268	0.164	0.094	0.079	0.0262	0.0180
	WITS 2012	0.182	0.123	0.090	0.275	0.164	0.096	0.069	0.0252	0.0172
	WITS 2087	0.173	0.124	0.089	0.262	0.173	0.100	0.079	0.0229	0.0153
	WITS 2191	0.179	0.128	0.089	0.265	0.168	0.102	0.068	0.0244	0.0162
	WITS 3156	0.178	0.123	0.095	0.269	0.165	0.100	0.070	0.0253	0.0173
	WITS 3314	0.185	0.124	0.092	0.266	0.167	0.097	0.069	0.0260	0.0174
	WITS 3316	0.175	0.124	0.091	0.272	0.171	0.102	0.065	0.0231	0.0154
	WITS 3360	0.181	0.129	0.094	0.251	0.171	0.101	0.072	0.0259	0.0168
	WITS 3364	0.180	0.128	0.095	0.271	0.155	0.097	0.074	0.0269	0.0185
	WITS 3369	0.176	0.127	0.096	0.258	0.167	0.099	0.077	0.0256	0.0172
	WITS 3372	0.174	0.126	0.086	0.273	0.169	0.103	0.070	0.0224	0.0151
	WITS 3388	0.176	0.127	0.094	0.267	0.164	0.104	0.068	0.0247	0.0167
	WITS 3448	0.182	0.129	0.091	0.264	0.165	0.098	0.072	0.0259	0.0173
	WITS 3664	0.181	0.124	0.100	0.268	0.157	0.094	0.075	0.0279	0.0192
	WITS 3529	0.179	0.130	0.094	0.258	0.166	0.098	0.074	0.0262	0.0174
	WITS 3606	0.177	0.123	0.093	0.273	0.166	0.093	0.074	0.0256	0.0175
	WITS 178	0.178	0.127	0.092	0.260	0.169	0.099	0.074	0.0251	0.0167
	WITS 188	0.181	0.130	0.092	0.256	0.165	0.098	0.078	0.0259	0.0173
	WITS 1472	0.193	0.134	0.083	0.258	0.161	0.102	0.070	0.0255	0.0166
	WITS 2097	0.174	0.124	0.098	0.267	0.163	0.099	0.076	0.0256	0.0176
	WITS 906	0.187	0.124	0.092	0.267	0.161	0.093	0.075	0.0271	0.0185
	UV S22	0.189	0.126	0.090	0.270	0.163	0.093	0.070	0.0269	0.0181
	UV S45	0.183	0.125	0.085	0.283	0.169	0.089	0.068	0.0245	0.0165
	UV S56	0.177	0.126	0.091	0.276	0.163	0.098	0.068	0.0248	0.0170
	UV S54	0.183	0.129	0.091	0.264	0.167	0.093	0.072	0.0266	0.0176
	UV S59	0.182	0.128	0.090	0.277	0.162	0.092	0.067	0.0264	0.0178
	UV S38	0.189	0.128	0.090	0.268	0.158	0.095	0.071	0.0272	0.0183
	UV S7	0.182	0.131	0.094	0.268	0.166	0.092	0.067	0.0275	0.0180

Species	Key	MC1	PP1	DP1	MC2	PP2	IP2	DP2	WS	GMI
	UV S26	0.186	0.127	0.086	0.273	0.165	0.092	0.071	0.0260	0.0175
	UV S17	0.174	0.125	0.088	0.271	0.173	0.097	0.073	0.0231	0.0153
	UV S52	0.181	0.131	0.091	0.264	0.164	0.100	0.069	0.0257	0.0170
	UNI-FI 4865	0.175	0.130	0.091	0.263	0.170	0.100	0.070	0.0247	0.0162
	UNI-FI 4887	0.171	0.126	0.094	0.261	0.171	0.103	0.073	0.0238	0.0158
	UNI-FI 4868	0.174	0.128	0.086	0.264	0.172	0.107	0.069	0.0219	0.0144
	UNI-FI 4880	0.173	0.131	0.091	0.268	0.162	0.102	0.074	0.0244	0.0167
<i>Pan troglodytes</i>	AMNH 89351	0.137	0.094	0.076	0.325	0.187	0.120	0.061	0.0092	0.0055
	AMNH 89353	0.139	0.094	0.071	0.325	0.189	0.119	0.063	0.0089	0.0052
	AMNH 89355	0.144	0.105	0.072	0.315	0.188	0.117	0.060	0.0112	0.0072
	AMNH 90189	0.148	0.094	0.064	0.322	0.182	0.125	0.065	0.0105	0.0065
	HMNH 11772	0.142	0.099	0.072	0.319	0.186	0.122	0.060	0.0104	0.0065
	NMW 25124	0.147	0.095	0.071	0.321	0.194	0.109	0.062	0.0090	0.0053
	NMW 13528	0.143	0.091	0.069	0.334	0.189	0.114	0.060	0.0068	0.0037
<i>Pan paniscus</i>	MRAC 29052	0.150	0.097	0.069	0.319	0.181	0.120	0.064	0.0121	0.0078
	MRAC 29044	0.161	0.101	0.059	0.335	0.175	0.106	0.062	0.0103	0.0062
	MRAC 27696	0.151	0.097	0.059	0.333	0.183	0.117	0.060	0.0083	0.0047
	MRAC 29042	0.142	0.095	0.077	0.317	0.178	0.118	0.072	0.0135	0.0088
	MRAC 29045	0.153	0.101	0.062	0.330	0.177	0.118	0.060	0.0102	0.0063
<i>Gorilla gorilla</i>	AMNH 81652	0.162	0.092	0.070	0.316	0.181	0.115	0.064	0.0127	0.0085
	AMNH 90194	0.161	0.101	0.063	0.314	0.184	0.111	0.066	0.0124	0.0079
	ZMB 18515	0.162	0.098	0.077	0.304	0.182	0.122	0.054	0.0128	0.0092
	NMW 792	0.157	0.120	0.060	0.309	0.179	0.112	0.063	0.0130	0.0084
<i>Pongo pygmaeus</i>	AMNH 18010	0.145	0.081	0.043	0.329	0.225	0.116	0.061	0.0010	0.0003
	AMNH 200898	0.144	0.087	0.049	0.310	0.216	0.125	0.070	0.0052	0.0025
	AMNH 90395	0.146	0.074	0.046	0.331	0.227	0.117	0.059	0.0004	0.0001
	HMNH 5154	0.146	0.085	0.048	0.332	0.207	0.124	0.059	0.0030	0.0012
	NMW 798	0.153	0.095	0.022	0.302	0.224	0.129	0.075	0.0036	0.0014
	NMW 795	0.150	0.074	0.040	0.341	0.218	0.123	0.053	0.0002	0.0000
	NMW 797	0.149	0.093	0.028	0.338	0.208	0.125	0.059	0.0019	0.0006
	NMW 800/B 5414	0.147	0.084	0.045	0.319	0.218	0.124	0.063	0.0034	0.0014
	NMW 654	0.149	0.088	0.042	0.333	0.220	0.114	0.052	0.0010	0.0003
	NMW 799	0.153	0.093	0.043	0.325	0.210	0.120	0.057	0.0038	0.0017
<i>Hylobates lar</i>	HMNH 35945	0.165	0.088	0.051	0.307	0.201	0.130	0.057	0.0075	0.0044
	HMNH 41421	0.155	0.092	0.045	0.293	0.206	0.151	0.058	0.0048	0.0025
	HMNH 41476	0.170	0.106	0.042	0.289	0.201	0.136	0.056	0.0065	0.0035
	HMNH 41504	0.174	0.092	0.044	0.309	0.205	0.122	0.055	0.0070	0.0039
	HMNH 41537	0.163	0.101	0.053	0.283	0.198	0.134	0.068	0.0077	0.0043
<i>Macaca mulatta</i>	UM-APC 319	0.190	0.114	0.061	0.290	0.182	0.102	0.061	0.0155	0.0100
	UM-APC 279	0.174	0.105	0.051	0.302	0.174	0.109	0.085	0.0129	0.0076
	UM-APC 318	0.169	0.112	0.054	0.295	0.177	0.115	0.078	0.0125	0.0076
	UM-APC 320	0.194	0.107	0.052	0.299	0.181	0.104	0.062	0.0134	0.0086
	UM-APC 224	0.180	0.103	0.056	0.314	0.182	0.102	0.064	0.0130	0.0084
	UM-APC 286	0.168	0.099	0.049	0.276	0.201	0.133	0.075	0.0074	0.0039
	UM-APC 231	0.198	0.118	0.061	0.239	0.199	0.118	0.066	0.0156	0.0085

Species	Key	MC1	PP1	DP1	MC2	PP2	IP2	DP2	WS	GMI
<i>Macaca fascicularis</i>	UM-APC 217	0.177	0.107	0.057	0.312	0.187	0.106	0.052	0.0115	0.0076
	UM-APC 281	0.186	0.109	0.059	0.298	0.181	0.100	0.067	0.0148	0.0096
	HMNH 35611	0.194	0.107	0.049	0.304	0.183	0.102	0.060	0.0127	0.0081
	HMNH 35612	0.178	0.116	0.055	0.293	0.183	0.110	0.065	0.0128	0.0079
	HMNH 35656	0.179	0.119	0.056	0.293	0.180	0.110	0.064	0.0135	0.0085
	HMNH 35658	0.189	0.110	0.052	0.297	0.183	0.104	0.065	0.0130	0.0081
	HMNH 35694	0.176	0.108	0.057	0.294	0.185	0.113	0.066	0.0124	0.0077
	HMNH 35729	0.175	0.108	0.062	0.298	0.191	0.103	0.063	0.0131	0.0084
	HMNH 35727	0.177	0.105	0.055	0.297	0.188	0.114	0.065	0.0115	0.0071
	HMNH 35736	0.189	0.107	0.058	0.284	0.178	0.120	0.064	0.0130	0.0084
	HMNH 35735	0.184	0.117	0.055	0.280	0.189	0.116	0.059	0.0121	0.0073
	HMNH 37349	0.175	0.118	0.057	0.281	0.192	0.110	0.066	0.0122	0.0073
	HMNH 37406	0.180	0.116	0.052	0.263	0.189	0.126	0.074	0.0111	0.0062
	HMNH 37407	0.180	0.117	0.044	0.280	0.183	0.122	0.073	0.0104	0.0058
	HMNH 37415	0.179	0.111	0.052	0.289	0.187	0.108	0.073	0.0123	0.0072
	HMNH 37416	0.182	0.112	0.055	0.287	0.184	0.109	0.071	0.0131	0.0079
	HMNH 57836	0.178	0.112	0.063	0.281	0.188	0.114	0.065	0.0130	0.0081
<i>Papio hamadryas</i>	AMNH 200847	0.201	0.109	0.067	0.310	0.161	0.090	0.063	0.0188	0.0134
	NMW 772	0.204	0.105	0.060	0.312	0.161	0.099	0.060	0.0165	0.0118
	NMW 825	0.202	0.103	0.062	0.301	0.160	0.093	0.078	0.0194	0.0133
	NMW 847	0.201	0.117	0.059	0.306	0.171	0.084	0.061	0.0173	0.0117
	NMW 28584	0.202	0.105	0.065	0.299	0.153	0.108	0.067	0.0188	0.0135
	NMW 43411	0.202	0.104	0.059	0.322	0.152	0.092	0.070	0.0173	0.0123
	NMW 22755	0.191	0.104	0.059	0.316	0.158	0.096	0.077	0.0174	0.0120
<i>Presbytis cristata</i>	HMNH 35666	0.155	0.097	0.048	0.304	0.211	0.125	0.060	0.0065	0.0035
	HMNH 35669	0.155	0.083	0.047	0.312	0.209	0.128	0.066	0.0059	0.0030
	HMNH 35640	0.155	0.085	0.046	0.310	0.214	0.124	0.066	0.0053	0.0026
	HMNH 35678	0.159	0.088	0.038	0.309	0.216	0.129	0.061	0.0046	0.0022
	HMNH 35675	0.150	0.085	0.044	0.302	0.217	0.139	0.063	0.0045	0.0022
	HMNH 35688	0.160	0.079	0.042	0.314	0.218	0.128	0.059	0.0037	0.0017
	HMNH 35683	0.159	0.082	0.045	0.314	0.220	0.125	0.055	0.0033	0.0015
	HMNH 37396	0.154	0.090	0.052	0.310	0.204	0.129	0.060	0.0069	0.0038
	HMNH 37665	0.155	0.074	0.045	0.320	0.216	0.132	0.058	0.0029	0.0012
<i>Sapajus apella</i>	AMNH 188037	0.167	0.137	0.092	0.226	0.182	0.119	0.077	0.0216	0.0130
	AMNH 95129	0.154	0.138	0.090	0.221	0.187	0.127	0.083	0.0183	0.0108
	AMNH 80169	0.144	0.139	0.092	0.226	0.188	0.120	0.091	0.0183	0.0109
	HMNH 61405	0.185	0.109	0.055	0.286	0.194	0.106	0.065	0.0122	0.0073
	YPM MAM 7577	0.174	0.126	0.065	0.261	0.193	0.126	0.055	0.0125	0.0073
	YPM MAM 6873	0.157	0.138	0.101	0.216	0.189	0.113	0.087	0.0224	0.0134
<i>Cebus albifrons</i>	AMNH 15485	0.169	0.130	0.093	0.232	0.174	0.111	0.092	0.0221	0.0144
	AMNH 23403	0.160	0.137	0.092	0.228	0.175	0.119	0.090	0.0204	0.0129
	AMNH 23401	0.157	0.139	0.105	0.204	0.185	0.118	0.093	0.0216	0.0128
	AMNH 23399	0.157	0.143	0.093	0.212	0.186	0.123	0.085	0.0202	0.0118
	AMNH 15484	0.149	0.131	0.097	0.231	0.176	0.119	0.097	0.0189	0.0122
	AMNH 200753	0.154	0.133	0.096	0.221	0.182	0.124	0.090	0.0190	0.0118
	AMNH 14017	0.156	0.141	0.094	0.212	0.191	0.121	0.085	0.0202	0.0118

Species	Key	MC1	PP1	DP1	MC2	PP2	IP2	DP2	WS	GMI
	AMNH 201434	0.155	0.139	0.088	0.226	0.188	0.127	0.077	0.0180	0.0105
	AMNH 201288	0.160	0.142	0.095	0.219	0.177	0.120	0.087	0.0213	0.0130
<i>Alouatta seniculus</i>	AMNH 23330	0.146	0.148	0.087	0.208	0.210	0.130	0.071	0.0162	0.0082
	AMNH 23334	0.150	0.141	0.089	0.202	0.204	0.134	0.079	0.0162	0.0086
	AMNH 23337	0.143	0.150	0.091	0.199	0.207	0.133	0.076	0.0169	0.0087
	AMNH 23335	0.155	0.140	0.091	0.206	0.207	0.126	0.075	0.0181	0.0097
	AMNH 23329	0.150	0.144	0.087	0.207	0.206	0.130	0.076	0.0169	0.0089
	AMNH 14660	0.140	0.148	0.094	0.203	0.204	0.130	0.081	0.0172	0.0091

Table S2. Result of principal component analysis (PCA) using the covariance matrix of hand proportions of the anthropoid species means.

	% var.	% var. cumulative	MC1	PP1	DP1	MC2	PP2	IP2	DP2
PC1	72.2	72.2	0.003	-0.397	-0.320	0.844	0.045	-0.016	-0.159
PC2	19.9	92.1	-0.551	-0.138	-0.176	-0.167	0.655	0.432	-0.055

Data is provided only for the axes accounting for most of the variance. The positive value represents the lengthening of the skeleton along the direction of PC while the negative value represents the shortening of the skeleton. Loadings with absolute values ≥ 0.3 are marked in bold. MC1, PP1 and DP1 are the abbreviations of metacarpal, proximal phalanx and distal phalanx for thumb. MC2, PP2, IP2 and DP2 are the abbreviations of metacarpal, proximal phalanx, intermediate phalanx and distal phalanx for forefinger.

Table S3. Bonferroni-corrected *post hoc* comparisons of the workspace size among anthropoid species.

	<i>Ho.</i> <i>sapiens</i>	<i>Pa.</i> <i>trogodytes</i>	<i>Pa.</i> <i>paniscus</i>	<i>G.</i> <i>gorilla</i>	<i>Po.</i> <i>pygmaeus</i>	<i>Hy.</i> <i>lar</i>	<i>M.</i> <i>mulatta</i>	<i>M.</i> <i>fascicularis</i>	<i>Pap.</i> <i>hamadryas</i>	<i>Pr.</i> <i>cristata</i>	<i>S.</i> <i>apella</i>	<i>C.</i> <i>albifrons</i>
<i>Pa. troglodytes</i>	0.000											
<i>Pa. paniscus</i>	0.000	1.000										
<i>G. gorilla</i>	0.000	0.162	1.000									
<i>Po. pygmaeus</i>	0.000	0.000	0.000	0.000								
<i>Hy. lar</i>	0.000	0.487	0.010	0.000	0.000							
<i>M. mulatta</i>	0.000	0.014	1.000	1.000	0.000	0.000						
<i>M. fascicularis</i>	0.000	0.006	1.000	1.000	0.000	0.000	1.000					
<i>Pap. hamadryas</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000				
<i>Pr. cristata</i>	0.000	0.000	0.000	0.000	0.126	1.000	0.000	0.000	0.000			
<i>S. apella</i>	0.000	0.000	0.000	0.002	0.000	0.000	0.000	0.000	1.000	0.000		
<i>C. albifrons</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.642	0.000	0.258	
<i>A. seniculus</i>	0.000	0.000	0.000	0.013	0.000	0.000	0.002	0.000	1.000	0.000	1.000	0.028

Statistical significance is set as 0.05, and the results of significant differences are marked in bold.

Table S4. Bonferroni-corrected *post hoc* comparisons of GMI among anthropoid species.

	<i>Ho.</i> <i>sapiens</i>	<i>Pa.</i> <i>trogodytes</i>	<i>Pa.</i> <i>paniscus</i>	<i>G.</i> <i>gorilla</i>	<i>Po.</i> <i>pygmaeus</i>	<i>Hy.</i> <i>lar</i>	<i>M.</i> <i>mulatta</i>	<i>M.</i> <i>fascicularis</i>	<i>Pap.</i> <i>hamadryas</i>	<i>Pr.</i> <i>cristata</i>	<i>S.</i> <i>apella</i>	<i>C.</i> <i>albifrons</i>
<i>Pa. troglodytes</i>	0.000											
<i>Pa. paniscus</i>	0.000	1.000										
<i>G. gorilla</i>	0.000	0.015	1.000									
<i>Po. pygmaeus</i>	0.000	0.000	0.000	0.000								
<i>Hy. lar</i>	0.000	0.341	0.005	0.000	0.002							
<i>M. mulatta</i>	0.000	0.085	1.000	1.000	0.000	0.000						
<i>M. fascicularis</i>	0.000	0.013	1.000	1.000	0.000	0.000	1.000					
<i>Pap. hamadryas</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000				
<i>Pr. cristata</i>	0.000	0.000	0.000	0.000	0.578	1.000	0.000	0.000	0.000			
<i>S. apella</i>	0.000	0.000	0.000	0.839	0.000	0.000	0.005	0.000	0.117	0.000		
<i>C. albifrons</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	1.000	0.000	0.179	
<i>A. seniculus</i>	0.000	0.000	0.283	1.000	0.000	0.000	1.000	1.000	0.000	0.000	1.000	0.000

Statistical significance is set as 0.05, and the results of significant differences are marked in bold.

Note S1. Explanation of choosing the in-hand manipulation as to assess the manipulative potential

Although anthropoid hands can perform many manipulative tasks, such as grasping a stone securely and then transforming to other position with the additional action from arm, the effect of the hands in the simple tasks is only comparable to the talons of birds, or even parallel-jaw grippers. Compared to the grippers, the remarkable capacity of anthropoid hands is embodied in the enhanced manual manipulation, especially the in-hand manipulation with the fingertip. The dexterous manipulation with fingertip is critical in the human evolutionary history for tool-related (tool-using and tool-making) behaviors [3] and in many daily activities for modern humans, e.g. writing with a pen or buttoning a shirt. Besides humans, many nonhuman anthropoid species are also capable of applying the pincer-grip efficiently and using their hands dexterously during extractive foraging and object manipulation [4,5]. The dexterous manipulation with fingertip allows the thumb and forefinger to pinch up a very fine object with the fingertip. In this case, the manipulated object can be moved subtly and quickly by actuating the smaller finger mechanisms, which is superior to the gross movements from a larger arm [6]. The in-hand manipulation of holding and manipulating objects with the fingertips between thumb and forefinger is perhaps the most complex fine motor skill [7,8]. Thus the complex task of in-hand manipulation with the fingertips between thumb and forefinger (figure 1) is chosen to estimate the manipulative potential in our model.

Note S2. Kinematic model of thumb-forefinger tip pinch

The thumb and forefinger are each simplified as a linkage model that imitates the corresponding hand skeletons (figure 2). The tips of thumb and forefinger pick up and hold a fine object like a needle or a matchstick. The motion of the tip-pinch is assumed to be limited in the dorsopalmar plane of hand (or namely, flexion-extension plane of the forefinger) [9], since the movement of the tip pinch in the daily activities is mainly located in the plane [10]. Under that assumption, the forefinger is modeled as a linkage with three degrees of freedom to allow the flexion and extension within the dorsopalmar plane, while ignoring the abduction and adduction away from the plane in the metacarpophalangeal (MCP) joint. An accurate kinematic model of the thumb is difficult to establish due to the complex mechanical form of joints [11,12]. And the morphology of the thumb joints has some differences between different anthropoid species [13]. But in order to set up a unified model to investigate the manipulation, the kinematic model of thumb should be simple and remain the primary and common movements among the anthropoid species as much as possible. We refer to the existing models [9,14,15] and establish the simplified kinematic model of thumb. The carpometacarpal (CMC) joint of thumb is modeled by a two degrees-of-freedom universal joint with the axes perpendicular to each other and to the thumb metacarpal, to act the abduction and adduction and the flexion and extension separately. The MCP and interphalangeal (IP) joints of the thumb are modeled as two hinge joints to allow the flexion and extension. The abduction and adduction of the thumb MCP joint is abandoned in our model because the movement rarely occurs isolated from that of the thumb CMC joint [12] and its range of motion is relatively small.

Note S3. Forward kinematics of the thumb and forefinger

For each digit, the position of its tip within the dorsopalmar plane $\mathbf{x}_i = [x_i \ y_i]^T$ can be expressed in

$$\mathbf{x}_i = H_i(\mathbf{q}_i), \quad (1)$$

where $i=1, 2$ separately represents the thumb and forefinger. $\mathbf{q}_i$ denotes the joint angle vector of the respective digit and H_i represents the kinematics equation. The positions of the tips of thumb and forefinger are computed in the Cartesian space of the base coordinate system, which is attached to the conjunct position of thumb and forefinger (figure S7). For the forefinger, the fingertip is always located

in the x-y dorsopalmar plane and the kinematic equation of the fingertip position $\mathbf{x}_2 = [x_2 \ y_2]^T$ can be simply expressed in

$$\mathbf{x}_2 = H_2(\mathbf{q}_2) = \begin{bmatrix} -L_{PP2} \cdot \sin(q_{MCP2}) - L_{IP2} \cdot \sin(q_{MCP2} + q_{PIP2}) - L_{DP2} \cdot \sin(q_{MCP2} + q_{PIP2} + q_{DIP2}) \\ L_{PP2} \cdot \cos(q_{MCP2}) + L_{IP2} \cdot \cos(q_{MCP2} + q_{PIP2}) + L_{DP2} \cdot \cos(q_{MCP2} + q_{PIP2} + q_{DIP2}) + L_{MC2} \end{bmatrix},$$

where $\mathbf{q}_2 = [q_{MCP2} \ q_{PIP2} \ q_{DIP2}]^T$.

Location of the tip of the thumb is firstly calculated in three-dimensional space. Different from the forefinger, the orientation of the metacarpal facet of the thumb on the trapezium is not parallel to that of the other fingers. The metacarpal of thumb has a divergence apart from the forefinger [13] and thumb metacarpal has also an axial rotation towards the palm owing to the concave arch of the carpal bones [16]. Thus the initial coordinate system $x_0-y_0-z_0$ (figure S7) in the CMC joint is set as a rotation of a divergent angle of 30° and following by the axial rotation of 45° from the base coordinate system [15]. The homogeneous coordinate transformations are used to convert location of the tip of the thumb to the base coordinate system. The homogeneous transformations are as follows

$$\begin{aligned} {}^{Base}_0\mathbf{T} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(30^\circ) & -\sin(30^\circ) & 0 \\ 0 & \sin(30^\circ) & \cos(30^\circ) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(45^\circ) & 0 & -\sin(45^\circ) & 0 \\ 0 & 1 & 0 & 0 \\ \sin(45^\circ) & 0 & \cos(45^\circ) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ {}^{CMC-FE}_0\mathbf{T} &= \begin{bmatrix} \cos(q_{CMC-FE}) & -\sin(q_{CMC-FE}) & 0 & 0 \\ \sin(q_{CMC-FE}) & \cos(q_{CMC-FE}) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^{CMC-AA}_{CMC-FE}\mathbf{T} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ \sin(q_{CMC-AA}) & \cos(q_{CMC-AA}) & 0 & 0 \\ -\cos(q_{CMC-AA}) & \sin(q_{CMC-AA}) & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ {}^{CMC-AA}_{MCP1}\mathbf{T} &= \begin{bmatrix} 0 & 0 & -1 & 0 \\ \sin(q_{MCP1}) & \cos(q_{MCP1}) & 0 & L_{MC1} \\ \cos(q_{MCP1}) & -\sin(q_{MCP1}) & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^{MCP1}_{IP1}\mathbf{T} = \begin{bmatrix} \cos(q_{IP1}) & -\sin(q_{IP1}) & 0 & 0 \\ \sin(q_{IP1}) & \cos(q_{IP1}) & 0 & L_{PP1} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ {}^{IP1}_{TIP1}\mathbf{T} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & L_{DP1} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

The homogeneous coordinate of the thumb tip $\mathbf{p}_1 = [x_1 \ y_1 \ z_1 \ 1]^T$ in the base coordinate system is expressed in

$$\mathbf{p}_1 = {}^{Base}_0\mathbf{T} \cdot {}^{0}_{CMC-FE}\mathbf{T} \cdot {}^{CMC-FE}_{CMC-AA}\mathbf{T} \cdot {}^{CMC-AA}_{MCP1}\mathbf{T} \cdot {}^{MCP1}_{IP1}\mathbf{T} \cdot {}^{IP1}_{TIP1}\mathbf{T} \cdot \mathbf{p}_{TIP1},$$

in which $\mathbf{p}_{TIP1} = [0 \ 0 \ 0 \ 1]^T$. According to the homogeneous coordinate in the three-dimensional

space, the position of thumb tip in the dorsopalmar plane $\mathbf{x}_1 = [x_1 \ y_1]^T$ is obtained by

$$\mathbf{x}_1 = H_1(\mathbf{q}_1) = \mathbf{M} \cdot \mathbf{p}_1,$$

with the constraint of $z_1 = 0$, where $\mathbf{M} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$, $\mathbf{q}_1 = [q_{CMC-FE} \ q_{CMC-AA} \ q_{MCP1} \ q_{IP1}]^T$.

Note S4. Derivation of the manipulative ellipsoid

Before establishing the measurement of manipulability, we first set up the kinematic relationship between the tips and joints of thumb and forefinger. Differentiating equation (1) with respect to time yields the kinematic relationship

$$\mathbf{v}_i = \mathbf{J}_i(\mathbf{q}_i) \dot{\mathbf{q}}_i \quad (2)$$

from the joint velocity vector $\dot{\mathbf{q}}_i$ to the fingertip velocity vector $\mathbf{v}_i$ through the Jacobian matrix $\mathbf{J}_i(\mathbf{q}_i)$, with $\mathbf{J}_i(\mathbf{q}_i) = \partial \mathbf{H}_i(\mathbf{q}_i) / \partial \mathbf{q}_i$. In our model, the fingertip of the thumb contacts the tip of forefinger at the same point without rolling and sliding the object. It indicates the fingertip velocity vectors of the thumb and forefinger are the same. Then the kinematic relationship of the two fingers can be compacted into

$$\begin{bmatrix} \mathbf{v} \\ \mathbf{v} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_1 & \\ & \mathbf{J}_2 \end{bmatrix} \dot{\mathbf{q}}, \quad (3)$$

where $\dot{\mathbf{q}} = \begin{bmatrix} \dot{\mathbf{q}}_1^T & \dot{\mathbf{q}}_2^T \end{bmatrix}^T$.

A common used measure of manipulative capability is the manipulability ellipsoid, which is proposed by Yoshikawa [17]. Here we postulate that the motion abilities of all joints of the fingers are the same. The joint angular velocities are considered with a constant unit norm, i.e., a unit sphere in the joint velocity space

$$\dot{\mathbf{q}}^T \dot{\mathbf{q}} = 1. \quad (4)$$

Inserting the expression in the kinematics equation (3) is given by

$$\dot{\mathbf{q}}^T \dot{\mathbf{q}} = \begin{bmatrix} \mathbf{v} \\ \mathbf{v} \end{bmatrix}^T \begin{bmatrix} \mathbf{J}_1 & \\ & \mathbf{J}_2 \end{bmatrix}^\dagger \begin{bmatrix} \mathbf{J}_1 & \\ & \mathbf{J}_2 \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ \mathbf{v} \end{bmatrix} = 1, \quad (5)$$

where the symbol $\dagger$ denotes the pseudo-inverse of the matrix. After expanding and simplifying, equation (5) is written as

$$\mathbf{v}^T \left[(\mathbf{J}_1 \mathbf{J}_1^T)^{-1} + (\mathbf{J}_2 \mathbf{J}_2^T)^{-1} \right] \mathbf{v} = 1, \quad (6)$$

which is called a velocity manipulability ellipsoid (the derivation of velocity manipulability ellipsoid for the dual cooperative digits refers to the works [18,19] but with some differences since the tips of the two digits are in the same position in our model). Equation (6) defines that the realizable fingertip velocity $\mathbf{v}$ is an ellipsoid in the task space when the joint velocity $\dot{\mathbf{q}}$ is a unit sphere in the joint velocity space. The directions of principal axes of the ellipsoid are given by the directions of the eigenvectors of $\left[(\mathbf{J}_1 \mathbf{J}_1^T)^{-1} + (\mathbf{J}_2 \mathbf{J}_2^T)^{-1} \right]^{-1}$ and the lengths of the axes by the square root of the eigenvalues.

Note S5. Numerical calculation of the workspace and global manipulation index (GMI)

The workspace of the tips of thumb and forefinger is computed by the numerical calculation method. The angle of each joint is discretized among its range of motion, with the step length of 1° . The forward kinematics (Note S3) is used to compute the positions of the tips at all the postures of the discretized joint angles. The locations of the tips of forefinger are in the two-dimensional Cartesian space (the dorsopalmar plane) while the locations of thumb tips are in three-dimensional space. The reachable positions of the tips of thumb within the two-dimensional dorsopalmar plane are obtained by retaining the locations with the condition of $|z_1| \leq (L_{MC1} + L_{PP1} + L_{DP1}) \cdot \pi / 180$. For the convenience of calculation, the dorsopalmar plane is also discretized into equally spaced grid points. The interval ε of grid points is set as 0.005 in both x and y directions. All the discretized locations of tips of thumb and forefinger within the dorsopalmar plane are attached to the grid points according to the nearest distance. The overlapping area of the fingertips between thumb and forefinger is the reachable workspace where the fingertip of thumb and forefinger can pick up and manipulate a small object. The size of the reachable workspace is expressed in

$$S_{ws} = \varepsilon^2 \cdot n,$$

where n represents the number of the reachable grid points in the overlapping area of the fingertips of thumb and forefinger.

We use the reciprocal of condition number to measure the dexterous manipulative ability in one posture. But the postures of thumb and forefinger have infinite possibility in a certain reachable point of the fingertip's workspace due to the redundant number of joints. We choose maximum value of the reciprocal of condition number as the dexterity (dex) among all the possible combinations of the postures between thumb and forefinger in each grid point of the dorsopalmar plane. And the global manipulation index is calculated by

$$GMI = \varepsilon^2 \cdot \sum_{i=1}^N dex_i,$$

in which dex_i represents the dexterity of the grid point i in the reachable workspace.

Note S6. Evolutionary history of the hand and brain about manipulative ability

The manipulative ability is doubtlessly highly refined in humans. The ability is traditionally considered to continuously evolve among anthropoids to become most pronounced in humans. Within anthropoids, some leaf-eating monkeys show relatively little interest in object manipulation [20]. They only have a few types of actions, such as picking up, mouthing and transporting the object. The great apes excel at object manipulation and can even do fairly precise manipulations in relation to the tool use. For instance, our closest relative, chimpanzees, can proficiently probe the nests with thin stems to obtain the termites, or use stones to crack open nuts [21,22]. Thus, in the aspect of actual manipulative behavior, the higher anthropoids show more complex manipulative tendencies and skills [23-25]. The manipulative ability may be a better survival skill, and many animal species have the evolutionary convergence towards the manipulative behavior [26,27].

Undoubtedly in the evolutionary history of anthropoids, a most distinctive trait is the tendency towards the development of brain [28]. Both human and ape brains exhibit specializations with respect to other anthropoid brains [29]. Ape specialization predominantly includes a striking increase in the cerebellum size [30]. Human specialization is not only shown in the enlargement of cerebellum but an overall larger brain in proportion to the body size [28], with the strengthening of the neural connectivity in the cerebral cortex [29,31]. The development of brain is benefit for the skilled hand use. The augmentation of cerebellum improves the functioning of motor-related cortical regions with which it is connected, such as motor and premotor cortex. And the elaboration of function in these cortex then offer apes much more planning capacity of complex movements than monkeys [29,30,32,33]. The enlargement of human brain improves the intelligence and knowledge to have possibility to acquire more control repertoires of undertaking and mastering a large of sophisticated tasks [34]. The brain enhancement in the evolutionary process is probably a primary cause of the elaboration of the actual manipulative behaviors among anthropoid species.

However, the hand itself does not coevolve with the brain to promote the strengthening of the manipulative skills together. From the aspect of manipulative potentials assessed by hand proportions, our phylogenetically near relatives, chimpanzees and orangutans have lower manipulative potentials than many monkeys, such as macaques, capuchin monkeys, and howler monkeys. Commonly, hand and brain are two most important organs for humans. The hand function depends on the interaction with the brain and the intension of the brain needs the hand to execute. The hand can be regarded as the extension of the brain and the development of hand is always accompanied by important changes in the brain [35]. The key to the evolution of brain intelligence lies in the peculiarities of the hand. Without our hands, there wouldn't have been any point in growing big brains. The great utility of tools also imposes a strong selective pressure on the hand and the brain to coevolve so as to maximum of the manipulative ability. Thus the hand and brain should be interdependent and co-evolutionary through our primate ancestors [36]. Nevertheless, the highly intelligent chimpanzees unexpectedly have lower potentials of manipulation than some Old World monkeys based on hand proportions. The hand does not coevolve with brain to facilitate the manipulative ability. It suggests that although object manipulation is a direct way of interacting with and changing the surrounding environment that contribute to survival [37], the evolutionary changes of the nonhuman anthropoid hands may not trend to better manipulation.

Note S7. Evolutionary explanation of the morphological changes of hand proportions

Human hand is free from locomotion behaviors and specially adjusts for a single purpose: the act of grasping and manipulating. The hand specialized adaptation gives humans a major evolutionary advantage and makes human hand have more manipulative skills [38]. Compared with human hand, it is generally considered that the hand morphologies of non-human anthropoids are derived by compromising the particular needs for locomotion and manipulation [39-43]. However, our results show that the changes of non-human anthropoid hands are not towards to better manipulation in a large extent. On the contrary, the evolutionary change of hand proportions is much more under selective pressure for locomotion [44-47].

From the aspect of the morphology of hand proportions, some great apes have lower manipulative potentials than the monkeys. The low manipulative potentials are a consequence of their relatively long fingers and short thumb. This morphological characteristic of the ape hands is most likely driven by locomotor constraints. Both monkeys and apes are adapted to the life in the tree. When moving quickly above the branch of trees, the small-sized monkeys use hand to grasp the branches and maintain the balance with tail [48]. But it is hard for apes to maintain the balance above the branch as an evolutionary loss of a tail and an increase of heavy body weight [39,49]. Instead, apes often adopt the suspensory locomotion under the branches to travel from tree to tree [39,50]. To adapt the suspensory behaviors, the fingers of the apes have to elongate so that to flex completely around the branches in a firm hook-like grasp, while a long thumb would hinder the locomotor way [49]. The morphological changes of ape hands are in order to adapt arboreal locomotion from tree to tree as much as possible. However, the selective pressure of locomotion is largely at the expense of manipulative skills. It extremely reduces the manipulative potentials for these ape hands. The poor manipulative potentials given by hand proportions directly determine that apes have much less dexterous manipulation than humans [41,51], although the apes have high brain intelligence and can greatly exploit the manipulative potential. Performance observations also confirm the weak manipulative skills of the apes. Albeit the chimpanzees and orangutans can frequently pick up objects with thumb tip against the index tip, no following manipulative movement between thumb and forefinger appear to take place to change the position of the object [41,52]. The changes in the orientation of an object are affected primarily by movements at the wrist and forearm joints [53]. The movement of the apes with the thumb and forefinger thus appear mechanical. It may be further inferred that manipulative abilities of great apes will never inevitably achieve human status some days if the locomotor constraints are still required.

Our functional model also demonstrates that the changes of hand proportions in some New World and Old World monkeys can lead to the enhancement of manipulative potentials. But the enhancement cannot be well explained by the selection pressure for manipulation, because the hand proportions in the monkeys have notably changed whereas the manipulative potentials have only slightly enhanced. The strengthening of manipulative potentials is probably just the by-product of an adaption of locomotion in these monkeys. The argument of manipulative potentials in the New World monkeys is mainly attributed to the shortening of metacarpal of finger and the lengthening of phalanges of thumb. The morphological changes are typically explained as an adaption for arboreal locomotion in a fine-branch milieu [54,55]. The New World monkeys always forage for the fruits and insects located among narrow branches of trees and shrubs, the evolutionary change of a relatively short palm enable them to more effectively grasp slender branches during the locomotion for foraging [56]. It can allow hands to wrap completely around the narrow branches, and thus be able to generate enough torques to counterbalance pithing and rolling of the body's center of mass on the thin and flexible arboreal supports [54]. With respect to the other Old

World monkeys, the changes of hand proportions in baboons result in the relative enhancement of manipulative potential by the proportionally shortening of phalanges of finger and the lengthening of metacarpal of thumb. The morphological changes are mostly likely as a response to the stresses caused by terrestrial digitigrade locomotion [47,57]. Different from the arboreal quadrupeds such as macaques, baboons are highly specialized terrestrial digitigrade monkeys. In the digitigrade locomotion, the palmar surfaces of phalanges make contact with the ground and the metacarpals are hyperextended [58,59]. The shortening of the phalanges of finger in baboon hands helps to bear the weight in the frequent terrestrial quadrupedalism [60]. Thus, the evolutionary change of proportions in both Old World monkeys and New World monkeys can be considered as an adaption of locomotion, with the by-product of the enhancement of manipulative potentials.

Note S8. The effect of locomotion on joint mobility

Whether humans or animals, the joint mobility is plastic and can be influenced by age, sex, and environment [61-64]. The newborn babies have great joint mobility while the degree of joint mobility diminishes rapidly during the childhood [61,62,65,66]. Such development of joint mobility is probably for the sake of stability [67]. The joint is immature and unstable around the time of birth and cannot bear large force. If there are large translational forces in parallel to the articular surface, these forces must be counteracted by ligaments or stabilizing muscle activity, redirecting the joint reaction force towards the articular surface [68]. For proper joint stability and preventing the possible damage like dislocation, the joint translations should be constrained either by compressing the head in the socket or by ligamentous structures. In the development of joint, the ligaments will become tight to increase the stability at the cost of the range of motion. The requirement of stability is more rigorous for nonhuman anthropoid hands than those of humans. The nonhuman anthropoid hands are under the extra constraints from locomotor behaviors as they are needed to participate in the quadrupedal locomotion. The locomotor behaviors result in largely compressive loads across the carpal and carpometacarpal joint due to ground reaction forces [69]. Thus, the quadrupedal monkeys are subjected vast compressive loads transmitted from the forelimb to the carpus. For stability, the range of motion of carpometacarpal joint is strictly constrained in these species. However, among the quadrupedal anthropoids, the great ape of chimpanzees and gorillas adopt a specialized form of quadrupedalism, terrestrial knuckle walk [70]. Their thumbs are not need to support the weight-bearing, and the mobility of carpometacarpal joint does not decrease too much compared to humans [71]. Thus, different patterns of locomotor behaviors lead to different joint mobility. In other words, the joint mobility can be regarded as a mirror of the patterns of locomotor behaviors [72].

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